

CS 214: Discrete Mathematics

2025-26 Semester II

Summary

Discrete mathematics is a self-contained course that covers a wide range of topics: a first introduction to proof, graphs and other structures, logic, set theory, counting and probability. No prerequisites are expected. The main focus will be on learning to write mathematics and making careful arguments.

Outcomes

- Familiarity with techniques of proof.
- Read, understand and write intermediate proofs.

Evaluation and Logistics

Homework-based questions (answered in the tutorial after HW due date): 20%

In-class quizzes (best two of three will be considered): 20%

Midterm exam: 30%

Final exam: 30%

Frequent TA office hours will be provided to assist with mathematical writing. Missed tutorial questions or in-class quizzes will not normally be made up; documented exceptional circumstances may be considered at the instructor's discretion.

Tentative Outline

1. Proofs (4 weeks)
2. Mathematical Foundations (1 week)
3. Structures (3 weeks)
4. Counting and Probability (3 weeks)
5. Number theory (1 week)

Resources

1. Class notes and resources provided
2. Discrete Mathematics, Rosen
3. Mathematics for computer science, Lehman, Leighton, Meyer. (Available [online](#)).